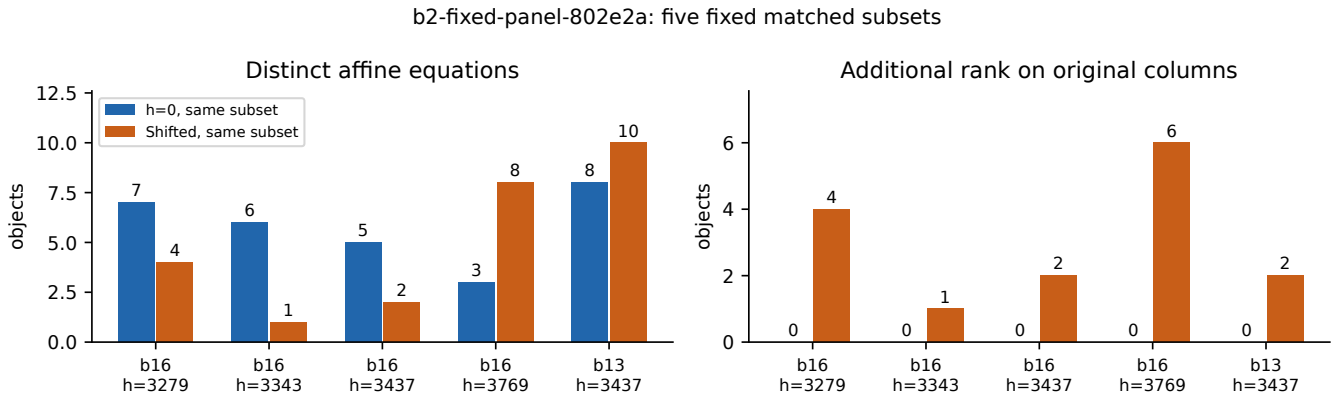
Matched shifted-cover collectors

Observations only. One fixed physical panel, with five matched pairs. The original two p4099 E curves, sixteen native atoms and fifteen projected unknown columns stay fixed. Each pair uses the same fourteen or fifteen active atoms for both constraints and residual acceptance, original64-source order, completed-zero policy and cubic pole(3,3) law. Only h changes. All five shifts were selected by the earlier membership-only survey, including every coverage-maximizing tie; no relation yield selected these cases.

Complete panel: True; collection attempts 15654912; separate replay attempts 15654912; unknown cells 0. The separate arithmetic checker reconstructs every branch/status/digest, both E and auxiliary point equations, native coefficients, affine ranks, complete kernels and old-kernel action. It is an implementation cross-check, not independent scientific review.

case	arm	atoms	rows	equations	new rank	old rank	combined	gain
b16-h3279	baseline	14	64	7	7	9	9	0
b16-h3279	shifted	14	37	4	4	9	13	4
b16-h3343	baseline	14	57	6	6	9	9	0
b16-h3343	shifted	14	7	1	1	9	10	1
b16-h3437	baseline	14	41	5	5	9	9	0
b16-h3437	shifted	14	20	2	2	9	11	2
b16-h3769	baseline	14	24	3	3	9	9	0
b16-h3769	shifted	14	65	8	7	9	15	6
b13-h3437	baseline	15	74	8	8	13	13	0
b13-h3437	shifted	15	98	10	10	13	15	2



The old matrices have rank9/15 on b16 and13/15 on b13. Gain is rank(M plus new rows)-rank(M) on the same original columns. Standalone rank or witness count is not additional information. Histories and each new row's old-kernel syndrome are retained. Counts are exact for these fixed cases; no curve-population or asymptotic inference follows.

Execution boundary

Run `b2-fixed-panel-802e2a`, exact source `c461dbb653a2c83c78d2a7614d387eef8f6324c6`. Frozen protocol canonical SHA256 `d55c58e0a3b0a5c54d0c7248288aa97b4ad8bee9731c473701c7e4e61108ebb1`. Exact committed inputs, release, interpreter/packages, real S3 gate, ownership/PIDs, host, stdout/stderr, per-source status blocks, accepted witnesses and native phase journals are beside the raw export. Lossless encodings retain original-byte hashes in archive-encodings. Data were committed/pushed before this report.

Exit 0, timeout False, source unchanged after True. Diagnostic wall 1870.714s, user CPU 1856.542s, system CPU 9.722s, peak RSS 221827072bytes. Shared host, wall_valid=false. Producer field/group counters omit helper primality/coefficient integer work, inline separate row verification, complete replay/elimination, hashing and storage accounting. No complete S, calibrated rate or attack-cost comparison is claimed.

- b16-h3279 baseline collection: complete, CPU 116.915s.
- b16-h3279 baseline verification: complete, CPU 61.883s.
- b16-h3279 shifted collection: complete, CPU 124.938s.
- b16-h3279 shifted verification: complete, CPU 78.275s.
- b16-h3343 baseline collection: complete, CPU 115.124s.
- b16-h3343 baseline verification: complete, CPU 71.336s.
- b16-h3343 shifted collection: complete, CPU 104.984s.
- b16-h3343 shifted verification: complete, CPU 64.941s.
- b16-h3437 baseline collection: complete, CPU 98.035s.
- b16-h3437 baseline verification: complete, CPU 61.723s.
- b16-h3437 shifted collection: complete, CPU 102.354s.
- b16-h3437 shifted verification: complete, CPU 64.559s.
- b16-h3769 baseline collection: complete, CPU 113.484s.
- b16-h3769 baseline verification: complete, CPU 61.613s.
- b16-h3769 shifted collection: complete, CPU 114.707s.
- b16-h3769 shifted verification: complete, CPU 72.274s.
- b13-h3437 baseline collection: complete, CPU 109.906s.
- b13-h3437 baseline verification: complete, CPU 68.329s.
- b13-h3437 shifted collection: complete, CPU 154.415s.
- b13-h3437 shifted verification: complete, CPU 102.181s.

Interpretation boundary and next use

Each pair is a finite relation-space comparison, with unchanged unknowns. Any pooled row-space or independent-target attachment is a separately labelled derived analysis. D1 retains the original rank13/9 matrices as its primary preprocessing, independently of these outcomes. Native success does not change an official research state. Small p101 controls preceded the freeze; an initial eight-atom nonzero-shift positive had no accepted row and was widened only as a marked software fixture, with the initial failure log preserved. No p4099 scientific input was changed by that correction.

Sources (internal/retrieved): prior B1 and complete cover-survey RESULTS.md; original relations/independent arithmetic; newly merged prym_leads_20261027 README at068985583; PKM2016 sections3.1–3.2, primary PDF digest a53030a6cde720c354aebfb65b1eb01e07766ce2d26e543d77f2516c0a49ef5b. Translated covers and complete-divisor duplicate explanations are existing work. Pole-complete unequal-infinity strata remain a separate unexecuted lead. Historical figures and records remain unchanged; this graph records only this panel. No novelty, E-isogeny, full-DLP speedup or general rank bound is asserted.